

ONLINE-SHOP APPLICATION DOCKER CONTAINERIZATION

This is the Notes for Hackathon Task to containerize the online-shop application

Step1: Create an EC2 instance/Virtual Machine

Step2: Clone the repository from github to your local server(EC2 instance)

Step3: Create a feature branch

Step4: Install Docker on your EC2 instance

Step5: Add ubuntu user to docker group

Step6: login to docker on ec2 instance using PAT(personal access token) from docker hub

Step7: Create a docker file with the necessary steps and packages as needed

Step8: Build the docker image

Step9: Run the Docker Container

Step10: Access the application from your web browser

Step11: Create a multi-stage Docker file

Step 12: Create a multi-stage image

Step13: Run the docker container with multi-stage Image

Step14: Access the application from your browser

Step15: Push your docker images to your docker hub

But while accessing the application from web browser I faced port mapping issue. Whereas in the documentation port number is mentioned as 3000 but as per the application requirements (vite.config.js) default port is 5173.

In this regard, to resolve the issue we can map the port while running the container using

Command: `docker run -d -p 3000:5173 --name online-shop online-shop:latest`

Or we can mention the port number explicitly in vite.config.js file.

Or we can directly access the application using the port 5173 in the browser directly since the default port for vite. Is 5173 as per the vite documentation

<https://vite.dev/config/server-options.html>

Image Creation Video Links:

https://drive.google.com/file/d/1h9B4KBNunswCYb3JtAb-2Vac1oMENSvY/view?usp=drive_link

https://drive.google.com/file/d/1skljwbusA7TPZV_FlgDPqqwv-U8TLEBY/view?usp=drive_link

Error Messages / Output Screen Shots:

```
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
online-shop   latest    ba03d3a5be11   37 seconds ago 256MB
node          18-alpine f6c173a701e1   4 weeks ago   127MB
```

```
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker run -d -p 3000:3000 --name online-shop online-shop:latest
3b4e2d24a4096cf6abcf23891e6253fed62151c4e0a95de69d968df42ac0fa0e
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
3b4e2d24a409   online-shop:latest "docker-entrypoint.s..." 2 minutes ago   Up 2 minutes   0.0.0.0:3000->3000/tcp, :::3000->3000/tcp   online-shop
```

```
ubuntu@ip-172-31-93-32:~/github/online_shop$ vi Dockerfile
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
3b4e2d24a409   online-shop:latest "docker-entrypoint.s..." 6 minutes ago   Up 6 minutes   0.0.0.0:3000->3000/tcp, :::3000->3000/tcp   online-shop
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker exec -it 3b4e2d24a409 bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "bash": executable file not found in $PATH: unknown
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker exec -it 3b4e2d24a409 sh
/app # curl http://localhost:3000
sh: curl: not found
/app # exit
```

```
ubuntu@ip-172-31-93-32:~/github/online_shop$ curl http://localhost:3000
curl: (56) Recv failure: Connection reset by peer
ubuntu@ip-172-31-93-32:~/github/online_shop$ ^C
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker logs 3b4e2d24a409
```

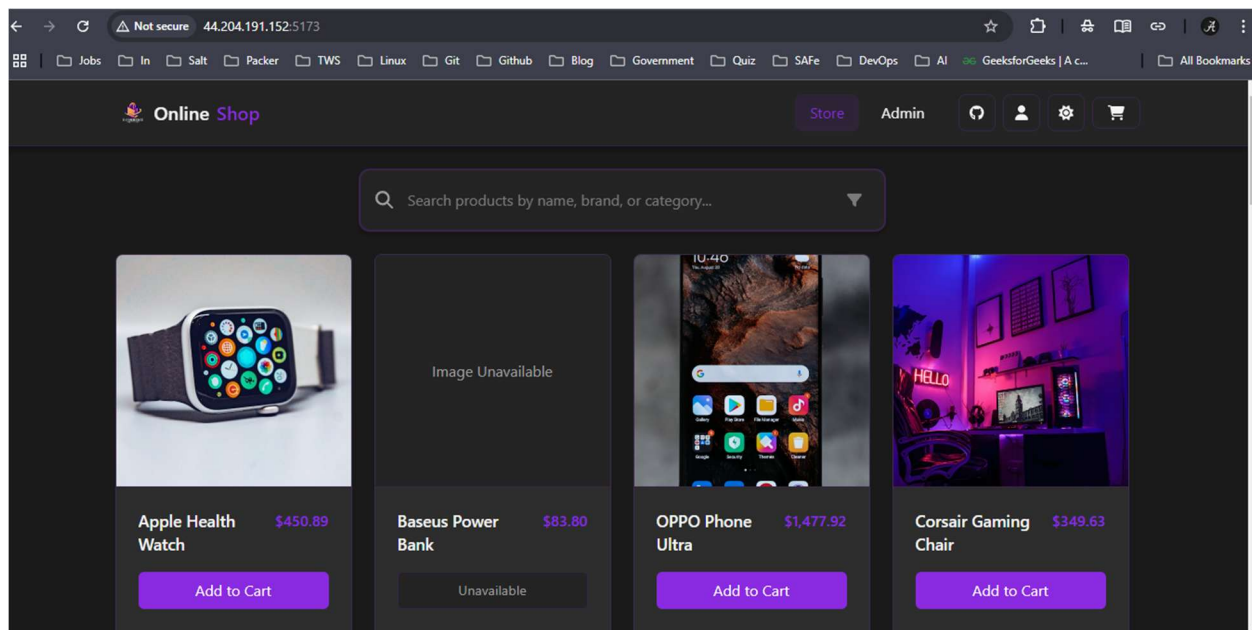
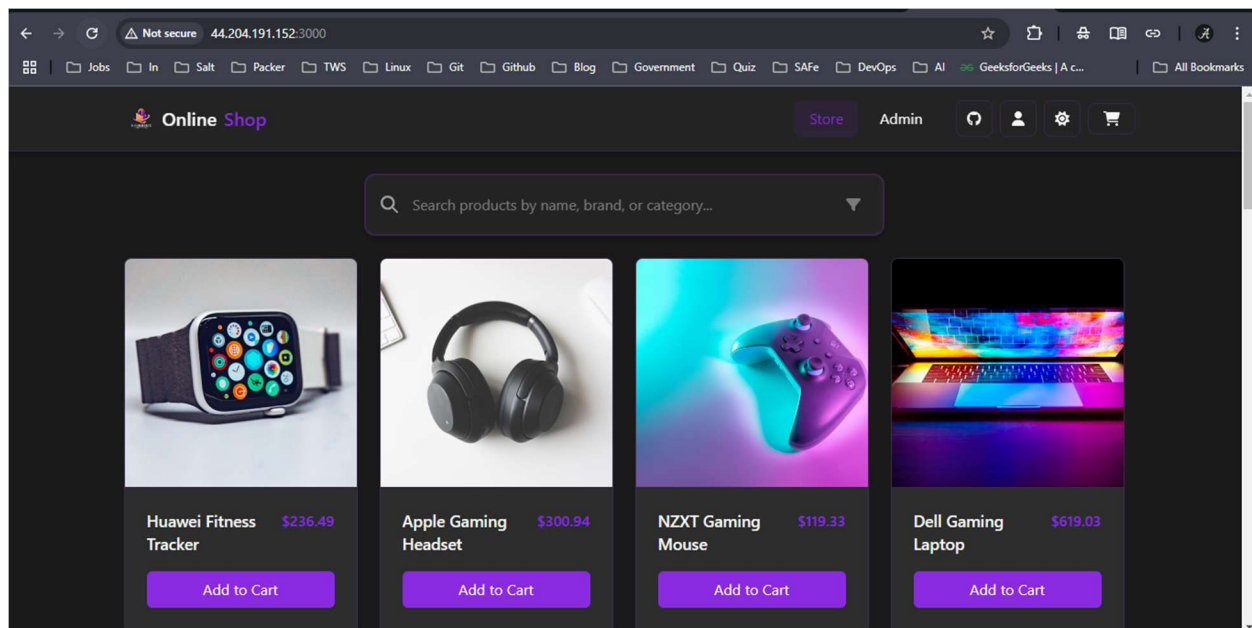
```
> online-shop@0.0.0 dev
> vite --host

VITE v5.4.14  ready in 244 ms

→ Local:   http://localhost:5173/
→ Network: http://172.17.0.2:5173/
```

```
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker stop 3b4e2d24a409
3b4e2d24a409
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker rm 3b4e2d24a409
3b4e2d24a409
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker run -d -p 5173:5173 --name online-shop online-shop:latest
48a3b81669779b7769a099ce6c68f1e77c31f93669f73b86b102d69a26eeaba3
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
48a3b8166977   online-shop:latest "docker-entrypoint.s..." 5 seconds ago   Up 4 seconds   3000/tcp, 0.0.0.0:5173->5173/tcp, :::5173->5173/tcp   online-shop
ubuntu@ip-172-31-93-32:~/github/online_shop$
```

```
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker run -d -p 3000:5173 --name online-shop online-shop:latest
e2d438b46e78965229d8cfe7c471c5596da60dbdf2817cfff24a4854ad3e27d2
ubuntu@ip-172-31-93-32:~/github/online_shop$
```



```
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
online-shop-multi-stage  latest             993d86de0f24       6 seconds ago      256MB
<none>               <none>             bac7acffd383       39 seconds ago     257MB
online-shop          latest             c932e1d6628a       9 minutes ago      257MB
node                 18-alpine          f6c173a701e1       4 weeks ago        127MB
ubuntu@ip-172-31-93-32:~/github/online_shop$
```

```

ubuntu@ip-172-31-93-32:~/github/online_shop$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                                NAMES
c5b4a45c83a0   online-shop-multi-stage:latest      "docker-entrypoint.s..." 4 seconds ago  Up 2 seconds  3000/tcp, 0.0.0.0:3000->5173/tcp, :::3000->5173/tcp  online-shop-mu
liti-stage
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker logs c5b4a45c83a0
> online-shop@0.0.0 dev
> vite --host

VITE v5.4.14 ready in 216 ms
→ Local:   http://localhost:5173/
→ Network: http://172.17.0.2:5173/
(!) Could not auto-determine entry point from rollupOptions or html files and there are no explicit optimizeDeps.include patterns. Skipping dependency pre-bundling.
ubuntu@ip-172-31-93-32:~/github/online_shop$

```

```

ubuntu@ip-172-31-93-32:~/github/online_shop$ docker images
REPOSITORY          TAG             IMAGE ID           CREATED           SIZE
online-shop-multi-stage  latest         993d86de0f24      36 minutes ago   256MB
<none>              <none>         bac7acffd383      37 minutes ago   257MB
online-shop          latest         c932e1d6628a      46 minutes ago   257MB
node                 18-alpine      f6c173a701e1      4 weeks ago      127MB
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker tag online-shop:latest ahamadb224/online-shop:latest
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker images
REPOSITORY          TAG             IMAGE ID           CREATED           SIZE
online-shop-multi-stage  latest         993d86de0f24      37 minutes ago   256MB
<none>              <none>         bac7acffd383      38 minutes ago   257MB
ahamadb224/online-shop  latest         c932e1d6628a      47 minutes ago   257MB
online-shop          latest         c932e1d6628a      47 minutes ago   257MB
node                 18-alpine      f6c173a701e1      4 weeks ago      127MB
ubuntu@ip-172-31-93-32:~/github/online_shop$ docker push ahamadb224/online-shop
Using default tag: latest
The push refers to repository [docker.io/ahamadb224/online-shop]
fc095414ab87: Pushed
398c669079ff: Pushed
4bee9d6fb38a: Pushed
03f0b2698171: Pushed
4692450629ab: Pushed
a9cfa7c9c02b: Mounted from library/node
77296e765ec2: Mounted from library/node
3766efcb8e44: Mounted from library/node
08000c18d16d: Mounted from library/node
latest: digest: sha256:9171618a237b1dbdc5e9436035f72d4b2a69478d413942ea6ae379c31d35548a size: 2205
ubuntu@ip-172-31-93-32:~/github/online_shop$

```

Commands Used:

1. mkdir "directory_name"
2. git clone "url repository"
3. cd "directory_name"
4. sudo apt-get update
5. sudo apt-get upgrade -y
6. sudo apt-get install docker.io -y
7. docker --version
8. sudo usermod -aG docker \$USER
9. cat /etc/group
10. docker images
11. vi Dockerfile
12. docker login -u "user_name"
13. docker build -t "image_name:tag_no".

14. docker run -d -p 3000:5173 --name "name_of_the_app" "image_name:tag_no"
15. docker ps
16. docker logs "container_id"
17. docker exec -it "container_id" sh
18. curl <https://localhost:3000>
19. docker stop "container_id"
20. docker rm "container_id"
21. docker build -f ./"file_name" -t "image_name" .
22. docker rmi "image_name:tag_no"
23. docker ps -a
24. git checkout -b "feature_branch_name"
25. git status
26. git add .
27. git commit -m "commit_message"
28. git push --set-upstream origin "feature_branch_name"
29. git tag "old_image_name:tag" "registry_name/image_name:tag"
30. docker push "registry_name/image_name:tag"

Improvements:

1. We can explicitly mention the port numbers in vite.config.js
2. We can make it two-tier applications where it uses data base server